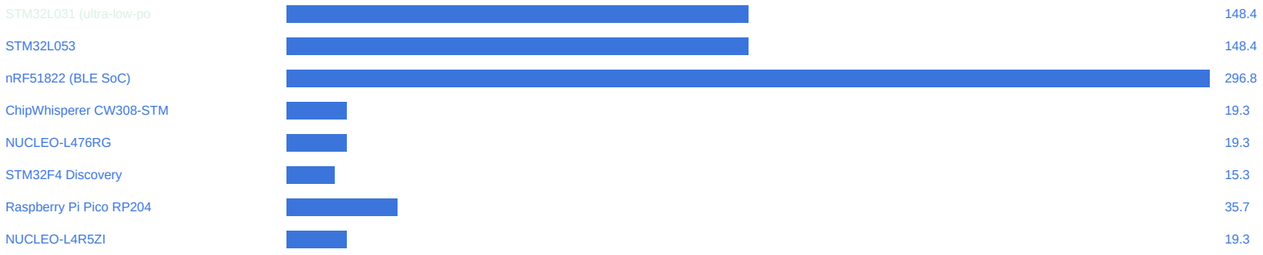
PQC-on-IoT Device Catalog and State Matrix

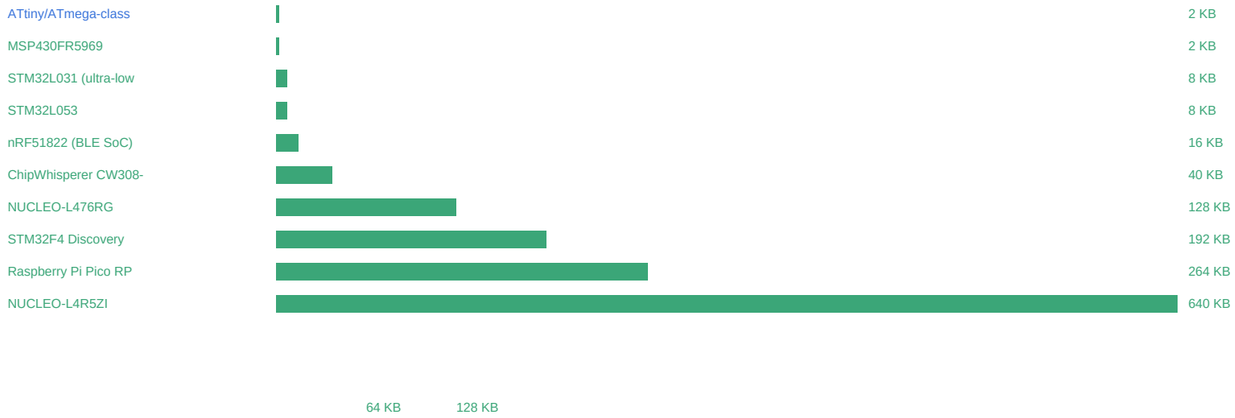
Generated 2026-11-01T06:51:32.738Z | Data 1.2.0 | Engine 1.2.0

Device	RAM	ML-KEM-512	ML-KEM-768	ML-KEM-1024	ML-DSA-44	ML-DSA-65	ML-DSA-87	SLH-DSA-128s	LMS H25
ATtiny/ATmega-class (8-bit AV	2 KB								
MSP430FR5969	2 KB								
STM32L031 (ultra-low-power)	8 KB								
STM32L053	8 KB								
nRF51822 (BLE SoC)	16 KB								
ChipWhisperer CW308-STM32F3	40 KB								
NUCLEO-L476RG	128 KB								
STM32F4 Discovery	192 KB								
Raspberry Pi Pico RP2040	264 KB								
NUCLEO-L4R5ZI	640 KB								

ML-KEM-512 execution time (ms)



RAM vs 64 / 128 KB workload floors



Projected values use the selected algorithm anchor; excluded architectures have no numeric projection.

Source: 01-DESIGN.md -4.2; M0+ @133 MHz